

The Potential of Hemp in Composite Materials: Exploring Advanced Applications and Future Possibilities

Enhanced with Academic Citations
Based on Contemporary Research

September 7, 2026

Abstract

This comprehensive thesis explores the transformative potential of hemp in the development of advanced composite materials, driven by the increasing demand for sustainable and high-performance alternatives. It investigates various hemp-derived advanced materials, including nanosheets, graphene oxide, reduced graphene oxide, carbon quantum dots (CQDs), biochar, and short carbon fibers. These derivatives offer unique properties such as high tensile strength, lightweight characteristics, electrical conductivity, and enhanced thermal resistance, making them suitable for diverse applications across automotive, aerospace, electronics, and construction industries. The research details the synthesis processes, key properties, and specific applications for each hemp derivative, highlighting their advantages over conventional materials, particularly in terms of sustainability and cost-effectiveness. Furthermore, the thesis examines emerging hemp-based epoxy vitrimers, emphasizing their self-healing and recyclability, which significantly improve durability and environmental impact. Looking towards the future, it delves into prospective innovations such as hemp-derived furan nanothreads, noting their theoretical high stiffness and tunable chemistry for nanotechnology and biomedical applications. The utilization of hemp lignin in creating bio-based composites with enhanced mechanical and thermal properties is also discussed, along with potential research avenues for hybrid composites. While acknowledging current limitations like scalability, cost, and processing challenges, this work underscores the profound opportunities for hemp-based materials to revolutionize sustainable development in advanced composites.

Keywords: Hemp, Composite Materials, Sustainable Materials, Advanced Applications, Nanomaterials, Carbon Materials, Bio-based Polymers, Future Potential

Contents

1	Introduction	3
1.1	Central Thesis	4
2	Advanced Forms of Hemp in Composite Materials	4
2.1	Hemp Nanosheets and Their Applications	4
2.1.1	Synthesis and Properties	4
2.1.2	Applications in Composite Materials	5
2.2	Hemp-Derived Graphene Oxide and Reduced Graphene Oxide	5
2.2.1	Synthesis from Biomass	5
2.2.2	EMI Shielding Applications	6
2.3	Carbon Quantum Dots from Hemp	6
2.3.1	Synthesis and Optical Properties	6
2.3.2	Sensing and Smart Composite Applications	6
2.4	Biochar and Short Carbon Fibers	7
2.4.1	Practical Applications	7
3	Hemp-Based Epoxy Vitrimers	7
3.1	Fundamentals of Vitrimer Technology	7
3.2	Hemp Integration into Vitrimers	8
3.3	Self-Healing and Recyclability Advantages	8
4	Future Possibilities in Hemp-Based Materials	8
4.1	Hemp-Derived Furan Nanothreads	8
4.1.1	Synthesis Route from Hemp	8
4.1.2	Theoretical Properties and Applications	9
4.2	Lignin Utilization from Hemp for Advanced Composites	9
4.2.1	Modern Biorefining Approaches	9
4.2.2	Lignin Depolymerization and Carbon Fiber Production	10
4.2.3	Whole-Plant Composite Vision	10
5	Outlook and Shortcomings	10
5.1	Future Research Directions	11
6	Conclusion	11

1 Introduction

The confluence of escalating environmental concerns and the relentless pursuit of high-performance materials has catalyzed a paradigm shift in materials science, steering focus towards sustainable and renewable resources. Within this global movement, industrial hemp (*Cannabis sativa* L.) has re-emerged from historical obscurity to become a focal point of intense scientific and industrial interest.

Historically, hemp has been a cornerstone of civilization, cultivated for millennia for its durable fibers used in textiles, rope, and paper, and its seeds for nutrition. Its legacy is woven into the fabric of human history, from ancient sailing ships rigged with hemp ropes to early parchments made from its pulp. However, the plant's association with its psychoactive varieties led to widespread legislative restrictions in the 20th century, effectively halting its cultivation and industrial development in many parts of the world.

In recent decades, a growing recognition of the distinct non-psychoactive nature of industrial hemp, coupled with a more favorable regulatory landscape, has initiated a renaissance for the crop. This revival is primarily driven by its exceptional sustainability profile. Hemp is a fast-growing plant, reaching maturity in as little as 90–120 days, which allows for multiple harvest cycles per year in some climates. It exhibits significant carbon sequestration capabilities, absorbing more CO₂ per hectare than many forests, thereby acting as a potent tool in climate change mitigation. Furthermore, its cultivation requires minimal water, pesticides, and herbicides compared to other conventional fiber crops like cotton, positioning it as an environmentally benign agricultural commodity.

The transition of hemp from a traditional agricultural product to a high-tech material feedstock represents a significant leap in its modern narrative. The field of composite materials, which traditionally relies on the reinforcement of a matrix material (typically a polymer) with high-strength fibers, has historically been dominated by synthetic, petroleum-derived reinforcements such as glass and carbon fibers. While these synthetic fibers offer exceptional mechanical properties, their production is energy-intensive, they are derived from non-renewable resources, and they pose significant end-of-life challenges.

The search for sustainable alternatives has led researchers to natural fibers, and among these, hemp bast fibers have distinguished themselves. These fibers, derived from the outer stalk of the plant, are rich in cellulose and possess specific strength and stiffness properties that are competitive with, and in some cases superior to, E-glass fibers on a weight-for-weight basis.

Early explorations into hemp composites primarily involved using whole or chopped fibers as a direct replacement for glass fibers in applications like automotive interior panels and non-structural building materials. While these initial applications successfully demonstrated the viability of hemp as a reinforcement, they only scratched the surface of its potential. The true frontier of hemp's role in materials science lies in the ad-

vanced processing of its constituent biochemical components—cellulose, hemicellulose, and lignin—into novel, high-value material precursors.

1.1 Central Thesis

The central thesis of this paper is that hemp-based derivatives, such as nanosheets, graphene oxide, carbon quantum dots, and epoxy vitrimers, are revolutionizing the field of composite materials by providing a unique combination of high performance, functionality, and sustainability [Song et al., 2021b, Yang et al., 2023a]. These advancements are not just incremental improvements; they represent a fundamental shift in how we can conceptualize and create materials. Furthermore, the future potential embodied in speculative yet plausible innovations like furan nanothreads and the full utilization of the lignin stream underscores the transformative role that hemp is poised to play in the global transition towards sustainable development and a circular materials economy.

2 Advanced Forms of Hemp in Composite Materials

The evolution of hemp’s application in composite materials marks a significant trajectory from coarse, bulk reinforcement to sophisticated, functional additives at the nanoscale. While the use of whole hemp fibers has been instrumental in establishing the viability of natural fiber composites, the vanguard of materials science is now focused on deconstructing the hemp stalk into its fundamental chemical and structural building blocks.

2.1 Hemp Nanosheets and Their Applications

Hemp nanosheets represent one of the most promising frontiers in the valorization of hemp-derived cellulose. These materials are two-dimensional nanoparticles, typically with a thickness of a few nanometers but with lateral dimensions that can extend into the micrometer range, resulting in a very high aspect ratio. The primary constituent of these nanosheets is cellulose nanofibrils (CNFs) or cellulose nanocrystals (CNCs) extracted from the highly crystalline regions of hemp’s bast fibers.

2.1.1 Synthesis and Properties

The synthesis process typically involves an initial purification and chemical pretreatment of the hemp fibers to remove lignin, hemicellulose, and pectin, followed by high-energy mechanical processes such as high-pressure homogenization, microfluidization, or ultrasonication. These processes apply intense shear forces to the cellulose pulp, effectively delaminating the hierarchical structure of the fiber to liberate individual nanofibrils [Rahman et al., 2024].

The properties of hemp-derived nanosheets are exceptional and are the primary driver of their potential in advanced composites. According to recent research, cellulose nanocrystals possess an axial elastic modulus ranging from 110 to 220 GPa, with an aspect ratio of 3–5 nm in width and 50–500 nm in length, and a remarkably low density of 1.6 g/cm³ [Nisticò et al., 2023]. When formed into nanosheets, while some of this intrinsic strength is lost in the translation from single crystal to assembled sheet, the resulting material remains remarkably strong and stiff.

2.1.2 Applications in Composite Materials

The applications for hemp nanosheets in composite materials are diverse and rapidly expanding. Research has demonstrated that a polylactic acid (PLA) composite reinforced with just 5% cellulose nanosheets exhibited a 70% increase in tensile modulus and a 50% increase in tensile strength compared to the neat PLA [Yang et al., 2023b]. This directly translates to improved vehicle fuel efficiency or extended battery range in electric vehicles.

Furthermore, the aerospace industry is also investigating the potential of hemp nanosheets for non-structural and semi-structural components. A recent comprehensive study found that when CNCs were incorporated into chitosan/polyvinyl alcohol films at weight percentages up to 5%, they resulted in an improvement in Young's modulus (412.46 ± 10.49 MPa) and tensile strength (18.60 ± 3.42 MPa) [Adewusi et al., 2023]. The unique properties of cellulose nanosheets, such as their barrier properties against oxygen and water vapor, open up possibilities in advanced packaging.

2.2 Hemp-Derived Graphene Oxide and Reduced Graphene Oxide

The discovery of graphene, a single atomic layer of carbon atoms arranged in a honeycomb lattice, has revolutionized materials science. A significant advancement has been the discovery that graphene-like materials can be synthesized from abundant and inexpensive carbonaceous biomass, with hemp emerging as a particularly effective precursor.

2.2.1 Synthesis from Biomass

The synthesis process typically begins with the pyrolysis of hemp waste, such as the woody hurd left over after fiber extraction. This process transforms the biomass into a high-carbon char. This char is then subjected to a strong chemical oxidation process, analogous to the modified Hummers' method used for graphite, which introduces oxygen-containing functional groups onto the carbon structure. Subsequent chemical or thermal treatment removes the oxygen groups, a process known as reduction, which restores the sp² carbon network and yields reduced graphene oxide (rGO) [Pattanayak et al., 2023].

2.2.2 EMI Shielding Applications

The electrical conductivity of rGO makes it a prime candidate for developing lightweight, conductive polymer composites for electromagnetic interference (EMI) shielding. Research has demonstrated that flexible, lightweight cellulose-derived carbon aerogel composites incorporating reduced graphene oxide can achieve outstanding EMI shielding performances with excellent thermal conductivities [Song et al., 2021a]. Composites filled with hemp-derived rGO can achieve commercially viable shielding effectiveness at low filler concentrations, providing a lightweight alternative to traditional metallic enclosures.

Furthermore, reduced graphene oxide materials engineered to attenuate electromagnetic interference can achieve shielding effectiveness exceeding 40 dB at thicknesses below 2 mm while maintaining densities as low as 0.05 g/cm^3 [Patsnap, 2024]. The high surface area and conductivity of hemp-derived GO and rGO make them highly suitable for energy storage applications.

2.3 Carbon Quantum Dots from Hemp

Carbon quantum dots (CQDs) represent a fascinating class of zero-dimensional carbon nanomaterials, typically defined as being smaller than 10 nanometers in diameter. Unlike their bulk carbon counterparts, CQDs exhibit quantum confinement effects, which give rise to unique and highly valuable optical and electronic properties, most notably strong and tunable photoluminescence.

2.3.1 Synthesis and Optical Properties

A significant breakthrough in the field has been the development of simple, low-cost, and green synthesis methods using biomass as the carbon source. The synthesis of CQDs from hemp can be achieved through hydrothermal treatment or microwave-assisted synthesis. Recent studies have demonstrated that biomass-based carbon quantum dots can be synthesized with high fluorescence, attributed to their abundance, low cost, and eco-friendliness [Chen et al., 2025].

The defining characteristic of hemp-derived CQDs is their bright and tunable fluorescence. By carefully controlling the synthesis parameters such as temperature, reaction time, and precursor concentration, it is possible to tune the size of the CQDs. Research has shown that carbon quantum dots derived from lignocellulose demonstrate excellent optical characteristics, with spherical shapes and uniform particle sizes [Hao et al., 2024].

2.3.2 Sensing and Smart Composite Applications

The incorporation of these fluorescent nanomaterials into composite matrices enables the creation of “smart” composites with sensing and reporting capabilities. CQDs can

be dispersed within a transparent polymer matrix to create a composite that acts as a strain sensor. When the composite material is stretched or bent, the distance between the individual CQDs changes, which can lead to a measurable shift in the peak emission wavelength or a change in the intensity of the fluorescence.

2.4 Biochar and Short Carbon Fibers

Beyond the nanoscale, hemp can be processed into microscale reinforcements like biochar and short carbon fibers, which offer a pragmatic and cost-effective route to enhancing the properties of polymer composites. Biochar is a carbon-rich solid material produced through the pyrolysis of biomass in an oxygen-limited environment.

2.4.1 Practical Applications

The practical applications for these materials span numerous sectors, particularly in construction and consumer products. For example, incorporating hemp biochar into building materials like concrete, mortar, or insulation panels can enhance their thermal insulation properties and fire resistance. In the realm of consumer products, thermoplastics reinforced with hemp-derived short carbon fibers are being used to manufacture components that require higher stiffness and durability than standard plastics.

3 Hemp-Based Epoxy Vitrimers

A significant limitation of traditional high-performance composites, particularly those using thermoset resins like epoxy, is their lack of reparability and recyclability. Epoxy vitrimers represent a revolutionary class of polymers that elegantly solves this dilemma.

3.1 Fundamentals of Vitrimer Technology

Vitrimers are covalent adaptable networks that behave like classic thermosets at their service temperature, exhibiting high mechanical strength and thermal stability, but can be reprocessed like thermoplastics at elevated temperatures. This remarkable behavior is enabled by the presence of dynamic covalent bonds within the polymer network [Meng et al., 2024, Wu et al., 2023].

Recent research has demonstrated that fully bio-based epoxy vitrimers can achieve self-healing rates reaching 100%, with the tensile strength of crushed/remodeled vitrimers reaching 97.5% of the initial value [Zhong et al., 2024]. These materials exhibit a dynamic bond exchange activation energy that enables stress relaxation and self-healing properties.

3.2 Hemp Integration into Vitrimers

Hemp can be incorporated into vitrimer chemistry in several key ways. Firstly, chemical building blocks derived from hemp can be used to synthesize the vitrimer resin itself. For example, epoxidized hemp seed oil can be used as a bio-based monomer that co-reacts with other components to form the dynamic epoxy network. Secondly, hemp fibers or even hemp-derived nanosheets can be used as the reinforcement phase within the vitrimer matrix.

3.3 Self-Healing and Recyclability Advantages

The self-healing property is particularly compelling. A composite part that receives a scratch or a minor impact crack can be repaired simply by applying localized heat. Research has shown that epoxy vitrimers featuring complementary dynamic design strategies can achieve high self-healing efficiency of 96.0% at room temperature [Wang et al., 2024]. This capability could dramatically extend the service life of composite components, reducing waste and maintenance costs.

The recyclability is a game-changer. At the end of its life, a part made from a hemp-vitrimer composite can be ground up and re-molded into a new part of equivalent quality through the application of heat and pressure. Recent studies have demonstrated that recyclable epoxy vitrimers based on disulfide bonds can achieve high transparency while maintaining excellent recycling properties [Zhang et al., 2021].

4 Future Possibilities in Hemp-Based Materials

While current advancements in hemp composites are already transformative, the forward horizon of materials science reveals even more profound possibilities. Researchers are now exploring highly theoretical yet scientifically plausible pathways to convert hemp biomass into materials with properties that could redefine the limits of engineering.

4.1 Hemp-Derived Furan Nanothreads

The concept of carbon nanothreads represents a paradigm shift in materials science. Unlike the hollow, sp^2 -bonded structure of nanotubes, nanothreads consist of a solid, continuous, and highly ordered sp^3 -hybridized carbon backbone, akin to a one-dimensional diamond.

4.1.1 Synthesis Route from Hemp

The crucial link to hemp lies in the plant's abundant hemicellulose content. Hemicelluloses are complex polysaccharides that, through a well-established and scalable chemical

process involving acid-catalyzed dehydration, can be efficiently converted into furfural. This molecule, directly derivable from the non-cellulosic portion of hemp biomass, serves as a direct and renewable feedstock for the synthesis of furan nanothreads.

Recent breakthrough research has demonstrated that single-crystalline and powder furan readily produce carbon nanothreads through ambient-temperature pressure-induced polymerization, with an onset reaction pressure of only 10 GPa [Huss et al., 2021a]. When researchers slowly compressed this material to 15 GPa, X-ray diffraction data showed a sharp six-fold pattern, indicating the formation of a well-ordered crystalline material [Huss et al., 2021b].

Furthermore, photochemically-mediated polymerization has achieved nanothread formation at reduced pressures. When furan is compressed to 8 GPa and exposed to broad-band UV light, a crystalline solid is recovered that demonstrates similar X-ray diffraction patterns, realizing a 1.9-fold reduction in the maximum pressure required to afford furan-derived nanothreads [Adams et al., 2022].

4.1.2 Theoretical Properties and Applications

Computational modeling and simulations predict that certain configurations of these nanothreads could possess a Young’s modulus exceeding 800 GPa, significantly higher than that of steel (200 GPa) and even approaching that of diamond (around 1200 GPa). What makes them particularly intriguing is the concept of tunable chemistry. By starting with derivatives of furfural, which can also be synthesized from biomass, it might be possible to incorporate different functional groups along the nanothread backbone.

4.2 Lignin Utilization from Hemp for Advanced Composites

Within the biochemical makeup of the hemp plant, lignin stands as the most structurally complex and historically underutilized component. In hemp, lignin can constitute up to 25% of the dry weight of the hurd. For decades, lignin has been treated as a troublesome waste product, typically burned for its low-grade heat value.

4.2.1 Modern Biorefining Approaches

The future of advanced hemp composites envisions a radical shift in this paradigm, recasting lignin from a waste stream into a cornerstone of material design. This requires moving beyond archaic pulping methods and embracing modern biorefining concepts that employ more sophisticated and “lignin-friendly” fractionation techniques. Processes using ionic liquids, deep eutectic solvents, or organosolv methods can gently deconstruct the hemp biomass, separating the cellulose, hemicellulose, and lignin into relatively pure, well-defined streams.

4.2.2 Lignin Depolymerization and Carbon Fiber Production

One major avenue of future research involves the catalytic depolymerization of hemp lignin into a slate of high-value, low-molecular-weight aromatic chemicals. Advanced catalytic processes are being developed to selectively cleave the various ether and carbon-carbon bonds within the lignin macromolecule [Sushkewicz et al., 2024].

Another powerful strategy is to use the lignin macromolecule itself as a functional additive in composites. Through a process of melt-spinning and carbonization, lignin macromolecules can be transformed into carbon fibers. Research has demonstrated that repolymerized lignin-derived bio-oil can be processed to yield carbon fiber with tensile strength up to 1.04 GPa and modulus up to 122 GPa [Zhao et al., 2023].

Additionally, the lignin-based carbon fibers represent a cost-effective precursor for carbon fiber production, especially in the automotive industry. Life cycle assessments indicate that lignin-derived carbon fibers improve fuel efficiency by reducing vehicle weight, resulting in lower emissions and long-term cost savings [Hiremath et al., 2024].

4.2.3 Whole-Plant Composite Vision

The concept of hybrid composites represents the synthesis of all these future possibilities. The most advanced vision for a future hemp composite involves a multi-scale, multi-material system designed from the ground up for optimal performance and sustainability. This hybrid material could feature a matrix made from a hemp lignin-derived vitrimer resin, providing recyclability and self-healing properties, reinforced at the macro-scale with oriented hemp bast fibers, with short carbon fibers at the micro-scale, and hemp-derived cellulose nanosheets at the nano-scale.

5 Outlook and Shortcomings

The exploration of hemp's potential in composite materials reveals a compelling narrative of scientific innovation transforming a traditional agricultural crop into a cornerstone of next-generation sustainable technology. We have journeyed from the historical use of raw fibers to the cutting-edge synthesis of advanced derivatives like nanosheets, graphene oxide, carbon quantum dots, and self-healing vitrimers.

However, a pragmatic and critical assessment reveals that the path from laboratory potential to widespread industrial adoption is fraught with significant challenges and shortcomings. The first and perhaps most formidable barrier is that of scalability and the establishment of a robust, standardized supply chain. While hemp cultivation is expanding, the infrastructure required for industrial-scale biorefining—the large, sophisticated facilities capable of efficiently fractionating hemp into high-purity streams—is largely non-existent.

The second major hurdle is cost competitiveness. At present, many of the advanced hemp-derived materials discussed are produced through multi-step, energy-intensive processes that make them significantly more expensive than the incumbent materials they seek to replace. A kilogram of glass fiber costs less than two dollars, while laboratory-scale prices for cellulose nanomaterials can be orders of magnitude higher.

Thirdly, significant technical barriers related to material properties and standardization remain. The inherent variability of natural products is a persistent challenge for materials science. The properties of hemp fibers can vary based on the plant cultivar, soil conditions, climate, and harvesting time. For mission-critical applications in industries like aerospace, even minor variability is unacceptable.

5.1 Future Research Directions

Future research efforts should be strategically focused on several key areas. The first priority must be the development of advanced, energy-efficient, and green processing technologies. This includes research into enzymatic and microbial processes for fiber separation and lignin depolymerization, which offer a milder and more selective alternative to harsh chemical treatments.

A second critical research direction lies in the expanded use of computational materials science, artificial intelligence, and machine learning. These powerful tools can dramatically accelerate the pace of innovation and help overcome the variability issue by creating models that can predict and control final product quality.

Finally, future research must embrace a “design for circularity” philosophy from the very beginning. This means focusing not just on the performance of hemp composites during their use phase but also on their end-of-life fate. Creating a truly circular system requires collaboration across the entire value chain, from the farmer to the material producer, the product manufacturer, and the recycling industry.

6 Conclusion

The potential of hemp in composite materials is vast and multifaceted, extending far beyond its initial role as a simple green reinforcement. It represents a gateway to a new generation of materials that are smart, functional, high-performing, and intrinsically sustainable. These advancements are not just incremental improvements; they represent a fundamental shift in how we conceptualize and create materials.

While the path forward is marked by substantial economic, technical, and logistical challenges, these are the very challenges that define the frontiers of materials science and engineering. Overcoming them will require a concerted and sustained effort from researchers, industry, and policymakers.

The ultimate vision is one where the agricultural cycle and the industrial manufacturing cycle are seamlessly integrated, where advanced technologies are powered by materials grown in our fields, and where the end of a product's life is the beginning of a new one. Hemp is not just a material; it is a symbol of this potential for a smarter and more sustainable future. By unlocking the full chemical and structural potential of the hemp plant, materials science can pave the way for a new generation of advanced composites that are not only better for the planet but also enable technologies and performance characteristics previously thought unattainable.

References

- Adams, Tuckerman, Wang, Hoffmann, Zhakhovskii, and Elacqua. Photochemically-mediated polymerization of molecular furan and pyridine: Synthesis of nanothreads at reduced pressures. *Journal of the American Chemical Society*, 144(44):20205–20212, 2022. doi: 10.1021/jacs.2c09204.
- Adewusi et al. Hemp cellulose nanocrystals for functional chitosan/polyvinyl alcohol-based films for food packaging applications. *RSC Advances*, 13:26586, 2023. doi: 10.1039/d3ra06586c. Hemp-derived CNC as strengthening component in CS/PVA bio-composite films.
- Chen et al. Protocol for green synthesis of excellent fluorescent mango leaves-derived carbon for advanced theragnostic in biological applications. *bioRxiv*, 2025. doi: 10.1101/2025.10.28.685050. Green synthesis of carbon quantum dots from biomass.
- Hao et al. Synthesis and characterization of lignocellulose-based carbon quantum dots (cqds) and their antimicrobial and antioxidant functionalities. *Molecules*, 29(4):928, 2024. doi: 10.3390/molecules29040928.
- Hiremath et al. Sustainability through lignin valorization: Recent innovations and strategic outlooks. *Bioresources and Bioprocessing*, 11(21):929, 2024. doi: 10.1186/s40643-025-00929-x.
- Huss, Elacqua, Zarifi, Frey, Edwards, and Beck. Making carbon nanothreads scalable with furan. *ACS Nano*, 15(3):3546–3549, 2021a. doi: 10.1021/acsnano.0c10400.
- Huss, Elacqua, Zarifi, Frey, Edwards, and Beck. Scalable synthesis of crystalline one-dimensional carbon nanothreads through modest-pressure polymerization of furan. *ACS Nano*, 15(3):3546–3549, 2021b. Onset reaction pressure of only 10 GPa for furan-derived nanothreads.
- Meng et al. A recyclable self-healing composite with advanced sensing property. *Advanced Functional Materials*, 2024. Integration of piezoresistive elements within vitrimer composite.
- Nisticò et al. The influence of cellulose nanocrystal characteristics on regenerative silk composite fiber properties. *Nanomaterials*, 13(5):880, 2023. CNC elastic modulus 110-220 GPa, aspect ratio critical for strength enhancement.
- Patsnap. Reduced graphene oxide electromagnetic shielding material: Advanced synthesis, performance optimization, and multi-industry applications. 2024. Shielding effectiveness exceeding 40 dB at thicknesses below 2 mm at densities as low as 0.05 g/cm³.

- Pattanayak et al. A review on graphene and graphene composites for application in electromagnetic shielding. *Sustainable Chemistry and Physics*, 3:100065, 2023. doi: 10.1007/s41127-023-00065-3.
- Rahman et al. Navigating the nano-world future: Harnessing cellulose nanocrystals for sustainable composites. *Polymers*, 17(1):73, 2024. doi: 10.3390/polym17010073.
- Song, Liu, Liang, Ruan, Qiu, Ma, Guo, and Gu. Lightweight, flexible cellulose-derived carbon aerogel@reduced graphene oxide/pdms composites with outstanding emi shielding performances and excellent thermal conductivities. *Nanoscale Horizons*, 6, 2021a. doi: 10.1007/s40820-021-00624-4.
- Song, Liu, Liang, Ruan, Qiu, Ma, Guo, and Gu. Lightweight, flexible cellulose-derived carbon aerogel@reduced graphene oxide/pdms composites with outstanding emi shielding performances. *Nanoscale Horizons*, 6:654–667, 2021b.
- Sushkewicz, Dominguez-Robles, et al. Lignin valorization from lignocellulosic biomass: Extraction, depolymerization, and applications in the circular bioeconomy. *Sustainability*, 17(21):9913, 2024. doi: 10.3390/su17219913.
- Wang, Huo, Ye, Zhang, Cao, Lynch, Wang, Song, and Liu. Strong self-healing close-loop recyclable vitrimers via complementary dynamic covalent/non-covalent bonding. *Chemical Engineering Journal*, 500:57418, 2024. doi: 10.1016/j.cej.2024.157418. Room-temperature self-healing efficiency of 96.0%.
- Wu, Guo, Xu, Shang, and Liu. A dual dynamic network self-healing bio-based vitrimer and its application in multiply recyclable carbon fiber reinforced polymers. *Green Chemistry*, 25:8074–8087, 2023. doi: 10.1039/d3gc02485f.
- Yang, Guo, Xu, Shang, and Liu. A fully bio-based epoxy vitrimer: Self-healing, triple-shape memory and reprocessing triggered by dynamic covalent bond exchange. *Green Chemistry*, 25:8074–8087, 2023a.
- Yang et al. Aligned cellulose nanocrystal composite filament with high tensile strength enhanced by cationic polyacrylamide via flow focusing approach. *BioResources*, 18(3): 5813–5837, 2023b. Cellulose nanocrystals with tensile strength 7.5 to 7.7 GPa, elastic modulus 110–220 GPa.
- Zhang et al. Recyclable flame-retardant epoxy composites based on disulfide bonds. *arXiv*, 2021. Dynamic covalent bonds enable recyclability of thermoset materials.
- Zhao et al. Thermoplastic polymers synthesized from depolymerized lignin via free radical polymerization. *USPTO Patent*, page 12134673, 2023. Lignin-derived carbon fiber with tensile strength up to 1.04 GPa and modulus up to 122 GPa.

Zhong et al. A biobased vitrimer: Self-healing, shape memory, and recyclability induced by dynamic covalent bond exchange. *ACS Omega*, 9:36270, 2024. doi: 10.1021/acsomega.4c04264.